AISC5000 Foundation of Machine Learning and AI

Homework 1

Due date: 11:59 pm, Mon, Oct 5, 2026

1. We have two observations represented by the feature vectors

$$x = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

   (a) Compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$. Then compute the cosine similarity

$$\cos\theta = \frac{x^\mathsf{T} q}{\|x\|_2 \, \|q\|_2}.$$

   (b) Let $S = \text{span}\{q\}$. Find the orthogonal projection $\widehat{x} = \text{proj}_S(x)$ and the residual $r = x - \widehat{x}$. Verify directly that r is orthogonal to q.

2. Consider a linear model $\widehat{y} = Xw$ with

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

   (a) Show that the columns of X are linearly independent. Compute $X^\mathsf{T}X$ and $X^\mathsf{T}y$.

   (b) Use the normal equation to find the least-squares estimator $\widehat{w}$. Compute the fitted vector $\widehat{y} = X\widehat{w}$ and the residual $r = \widehat{y} - y$. Verify the orthogonality condition $X^\mathsf{T}r = 0$.

   (c) Find the eigenvalues of $X^\mathsf{T}X$ and then the singular values of X.

3. For a one-parameter model, define

$$g(w) = we^w, \qquad L(w) = \frac{1}{2}\big(g(w) - 2\big)^2.$$

   (a) Use the product rule and chain rule to compute $L'(w)$. Evaluate $L'(0)$ and perform one gradient-descent update from $w^{(0)} = 0$ with learning rate $\eta = 0.1$.

   (b) Compute $L''(0)$ and the second-order Taylor polynomial of L about $w = 0$. Use it to approximate $L(0.2)$ and compare with the exact value (to four decimal places).

4. Let

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

and consider the regularized least-squares loss

$$J(w) = \frac{1}{2} \|Xw - y\|_2^2 + \frac{1}{2} \|w\|_2^2 .$$

(a) Compute $\nabla J(w)$ and the Hessian matrix $H_J(w)$.

(b) Find the critical point w^*. Use the Hessian to explain it is the unique global minimizer.

(c) Starting from $w^{(0)} = 0$, perform one gradient-descent step with $\eta = \frac{1}{6}$. Compute $J(w^{(0)})$ and $J(w^{(1)})$. Then approximate $J(w^{(1)})$ from the second-order multivariable Taylor formula of $J(w)$ at $w^{(0)}$.